



Teacher Code :

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SPECIALIST MATHEMATICS UNIT 2

Tech-Free

Test theme: SELBY'S CASTING CRISIS

Reading Time: 5 minutes

Writing time: 55 minutes

QUESTION AND ANSWER BOOK

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
6	6	33

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

**Question 1** (4 marks)

Selby enters the Crown Casino, clutching a single 50-cent piece. He sees Shaz, the local homeless person, muttering about the unit circle. In a rare moment of pity, Selby tosses her the 50 cents. 'It's all about the quadrants, kid,' she whispers. Selby heads to the high-stakes poker table. The dealer is Mr Tres, the annoying kid Selby can't stand. Mr Tres deals a hand where the 'Pot Odds' are determined by the variable x

$$\text{Given that } \sec(x) = -\frac{3}{2} \text{ and } \pi < x < \frac{3\pi}{2}.$$

a. Show that $\tan(x) = \frac{\sqrt{5}}{2}$

1 mark

b. Mr Tres giggles, 'If you want your payout, you need the value of $\tan(2x)$.' Calculate it.

1 Mark

- c. The 'Partial Fraction Dream' occurs when Selby notices the casino's ledger is written in identities. To prove Mr. Tres is cheating, Selby must verify the following:

$$\frac{1}{1-\sin(x)} + \frac{1}{1+\sin(x)} = 2\sec^2(x)$$

2 Marks

Question 2 (7 Marks)

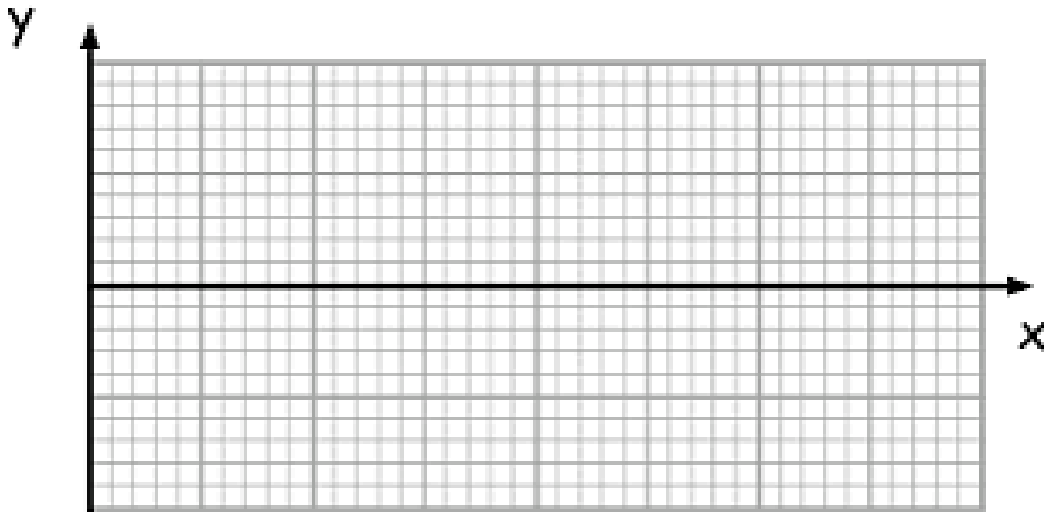
Selby wins big! He grabs his winnings and a single daisy from the casino vase, a symbol of his success. He sprints to his bike, but Mr. Tres has alerted the 'Trig-Police'. Selby pedals furiously.

- a. His left pedal's height follows $f(x) = \sqrt{2}\sin(x)$, and his right pedal follows $g(x) = \sin(x) - \cos(x)$. Selby's bike computer glitches. To recalibrate his speed, he must express $h(x) = f(x) + g(x)$ in the form $R\sin(x - \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Find the exact value of R and show that $\tan(\alpha) = \sqrt{2} - 1$.

3 Marks

- b. Selby takes a sharp turn. The path is only safe where $f(x) \geq g(x)$. On a single set of axes, sketch $y = f(x)$ and $y = g(x)$ for $x \in [0, \pi]$, clearly labeling the point of intersection

3 Marks



Working out:

- c. Find the exact x -coordinate of the intersection in the domain $x \in [0, \pi]$

1 Mark



Question 3 (6 Marks)

Selby reaches the pier. He sits in a field of daisies, clutching the one he stole from the casino.

He begins to assemble his rod. A composite inverse function models the tension in the line.

- a. Find the implied domain and range for the function $h(x) = 2\arcsin(2x - 1) + \pi$

2 Marks

- b. As Selby casts, the lure sinks to a depth defined by $d = \tan(2\arcsin(-\frac{3}{5}))$.

Calculate this exact depth

2 Marks

- c. A mysterious voice (likely Jimmy from the bushes) whispers that the perfect cast occurs when $x = \cos(\frac{\pi}{5})$. Using the fact that $\cos(\frac{\pi}{5}) = \frac{1+\sqrt{5}}{4}$, show that this value of x is a root of the equation $4x^2 - 2x - 1 = 0$

2 Marks

Question 4 (6 Marks)

Suddenly, Mr Fox, the final boss who has never gotten a math question wrong, appears. He doesn't want to fish; he wants to play Uno. He places a 'Draw 4' card down. 'Prove these identities, Selby, or draw the whole deck.'

a. Prove: $\frac{\sin(3\theta)}{\sin(\theta)} - \frac{\cos(3\theta)}{\cos(\theta)} = 2$

3 Marks

- b. Mr Fox places a 'Reverse' card. He demands that Selby find all solutions $\theta \in (0, \pi]$ to the equation $\cos(3\theta) = \cos(2\theta)$.

3 Marks

Question 5 (5 Marks)

The game of Uno ends in a draw. Mr Fox respects Selby's intellect and allows him one final cast. The trajectory is governed by a reciprocal expansion.

- a. Prove the identity: $\tan\left(\frac{x}{2}\right) + \cot\left(\frac{x}{2}\right) = 2\csc(x)$

2 Marks

- b. Selby's lure will hit a fish if he can solve $\tan\left(\frac{x}{2}\right) + \cot\left(\frac{x}{2}\right) = 4$ for $x \in [0, \pi]$.

Find the exact solution

3 Marks

Question 6 (5 Marks)

Mr Fox smirks. 'One last thing, Selby. A true Specialist Mathematician sees the hidden symmetry.' He points to the ripples in the water

a. Show that $\frac{1+\cos(2x)+\sin(2x)}{1-\cos(2x)+\sin(2x)} = \cot(x)$

3 Marks

b. Find all solutions for $x \in (0, 2\pi)$ such that $\frac{1+\cos(2x)+\sin(2x)}{1-\cos(2x)+\sin(2x)} = \sqrt{3}$

2 Marks

END OF TEST**CHECK YOUR WORKING, OR MR FOX WINS YOUR BIKE!**